



Of Chips And Sockets

Craving to upgrade to a Socket LGA 1156 processor? You'll need a supporting motherboard. Here's a look at the boards that support Intel's latest CPU

Rohan Naravane

rohan.naravane@thinkdigit.com

Last month, in our extensive motherboard test, we saw a variety of boards for both Intel and AMD platforms. From Intel, the mid-range performance segment comprised of boards having the P45 chip-set for Core 2 series of processors while the high-performance ones were the X58 chip-sets for the Core i7. After Core i7, Intel has planned to continue with the Core "i" moniker for future CPUs.

And then there were two sockets...

For a long time, the LGA 775 was the mainstream socket design used by all the Core 2 series processors. Currently, the LGA 1366 socket is what the high-end Core i7-9xx series sit on. The platform is the X58 chipset that we tested last month. Two months ago, Intel gave the world the Core i5 processor. A newer LGA 1156 socket was developed, which is compatible with all the Core i5 and the Core i7-8xx series processors. This socket will be based on the P55 chipset.

To point out a few differences between the Core i5 and the Core i7 — first, the Core i5 will not feature hyper-threading. Second, it will not support tri-channel memory; only dual-channel like the Core 2 platform. Lastly, it will also not feature Intel's point-to-point interconnection system called QPI (QuickPath Interconnect). But the Core i5 brings performance improvements over the Core 2 series being based on the mighty Nehalem architecture. Thus, we have the Core i7 as the performance king, while the Core i5 as the

mid-range performer in Intel's line-up. Thus Intel plans to do away with the LGA775 socket associated with all the Core 2 processors. There will be a Core i3 in the future to cater to the entry-level segment.

A quick brief on the major differences between the predecessor (the P45) and the successor (the P55); first, we have the memory controller integrated into the processor on the P55. Second, (and this is a new achievement for any platform), which is the connectivity of the available 16 lanes of PCI Express 2.0 to the

